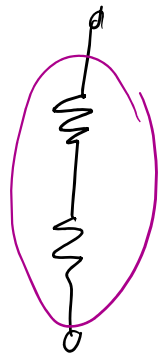
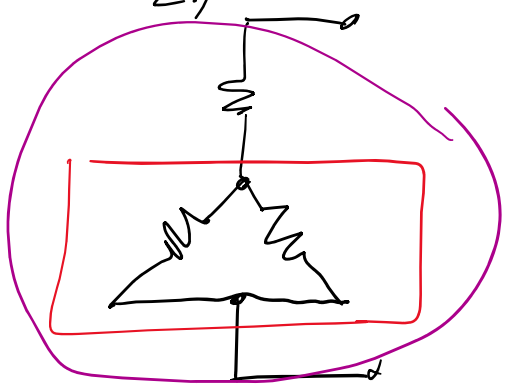
Lecture 6

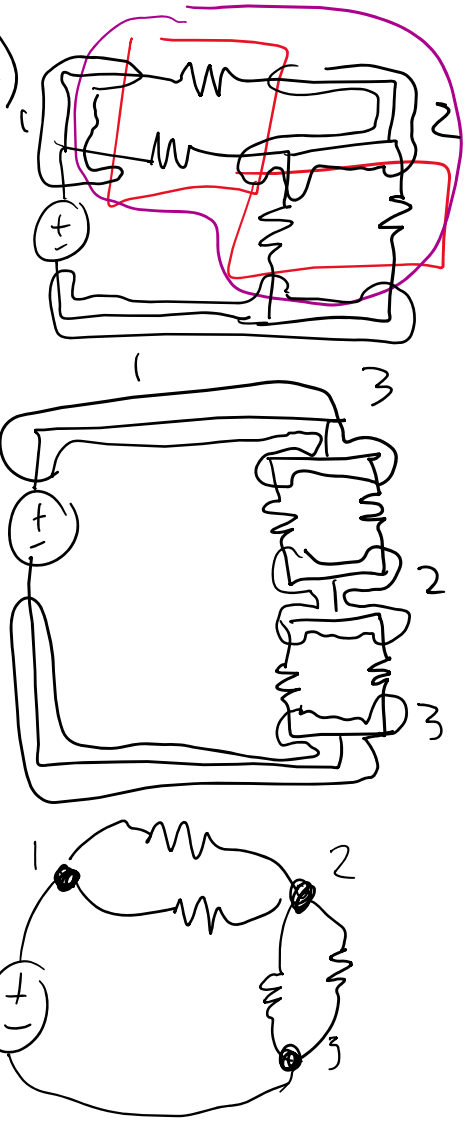
1.)



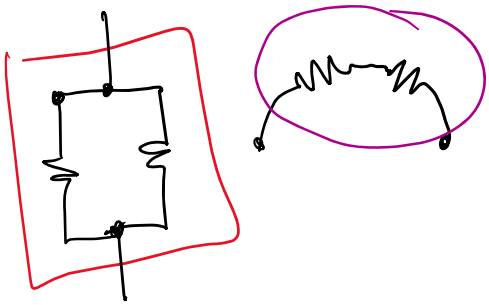
2.)



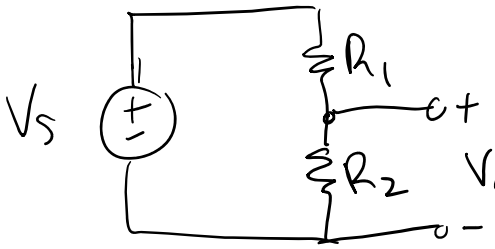
3.)



* Circle the series connections
* Box in the parallel

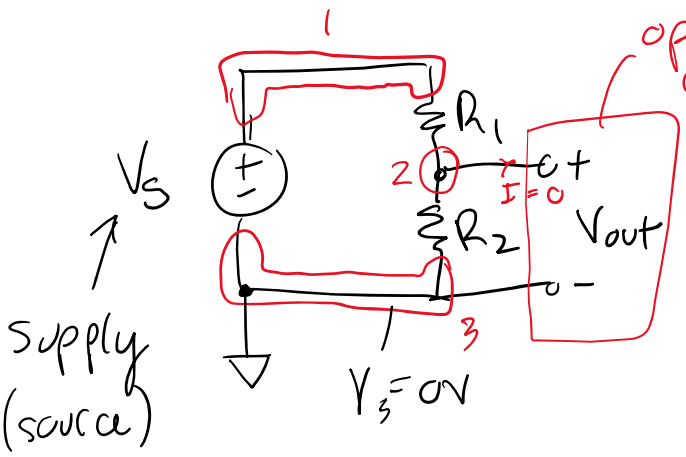


Voltage Dividers



(Vout is the $\overbrace{\text{output voltage}}^{\text{what we are interested in}}$ across R_2)

Voltage Dividers



open circuit $R=\infty$ what we are interested in

$I = \frac{V}{\infty} = 0$

(V_{out} is the output voltage across R_2)

Voltage
- we know voltage at node 1 and node 3.
- find Node 2

Node 2

$$V_2 = V_{out}$$

$$\frac{V_2 - V_s}{R_1} + \frac{V_2 - 0}{R_2} = 0$$

$$\frac{V_{out} - V_s}{R_1} + \frac{V_{out} - 0}{R_2} = 0$$

$$\frac{V_{out}}{R_1} - \frac{V_s}{R_1} + \frac{V_{out}}{R_2} = 0$$

$$\left(\frac{1}{R_1} + \frac{1}{R_2}\right)V_{out} - \left(\frac{1}{R_1}\right)V_s = 0$$

$$\left(\frac{1}{R_1} + \frac{1}{R_2}\right)V_{out} = \left(\frac{1}{R_1}\right)V_s$$

$$\left(\frac{1}{R_1}+\frac{1}{R_2}\right)V_{out} = \left(\frac{1}{R_1}\right)V_S$$

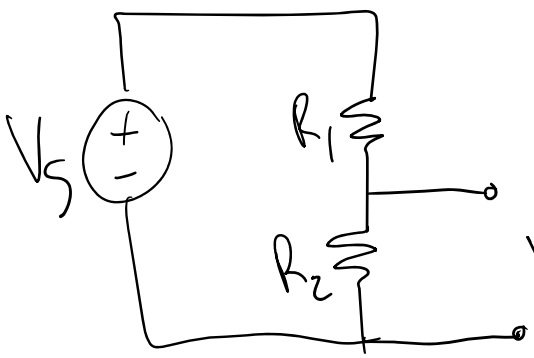
$$\left[\frac{1}{R_1}\left(\frac{R_2}{R_2}\right)+\frac{1}{R_2}\left(\frac{R_1}{R_1}\right)\right]V_{out} = \frac{1}{R_1}V_S$$

$$\left(\frac{R_2+R_1}{R_1R_2}\right)V_{out} = \frac{1}{R_1}V_S$$

$$V_{out} = \left(\frac{\cancel{R_1}R_2}{R_1+R_2}\right)\left(\frac{1}{\cancel{R_1}}\right)V_S$$

$$\boxed{V_{out} = \frac{R_2}{R_1+R_2}(V_S)}$$

→ Voltage divider is a "step-down" circuit



$V_S = 120\,V_{DC}$ → 110 ⇔ 120

$V_{out} = 1\,V$

$$1V = \frac{R_2}{R_1+R_2}(120V)$$

let $R_1 = 100k$

* we want a low current so power loss in Resistors is Low → 1mA

$$\frac{120}{R_1+R_2} = 1mA$$

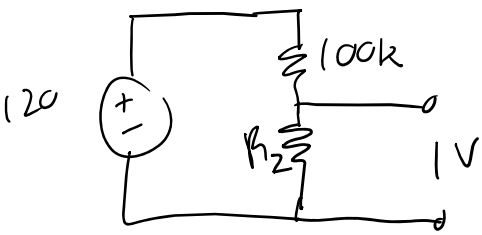
$$120k\Omega = \frac{120}{1mA} = (R_1+R_2) \Rightarrow$$

$V_S = 120V$

$I_S = 1mA \rightarrow$ we set this

$R_1 + R_2 = 120k\Omega$

let $R_1 = 100k\Omega$



$$V_{out} = \left(\frac{R_2}{R_1 + R_2}\right) V_S$$

$$1V(100k + R_2) = 120(R_2)$$

$$100k + R_2 = 120 R_2$$

$$100k = 120 R_2 - R_2$$

$$100k = (119) R_2$$

$$\frac{100k}{119} = R_2$$

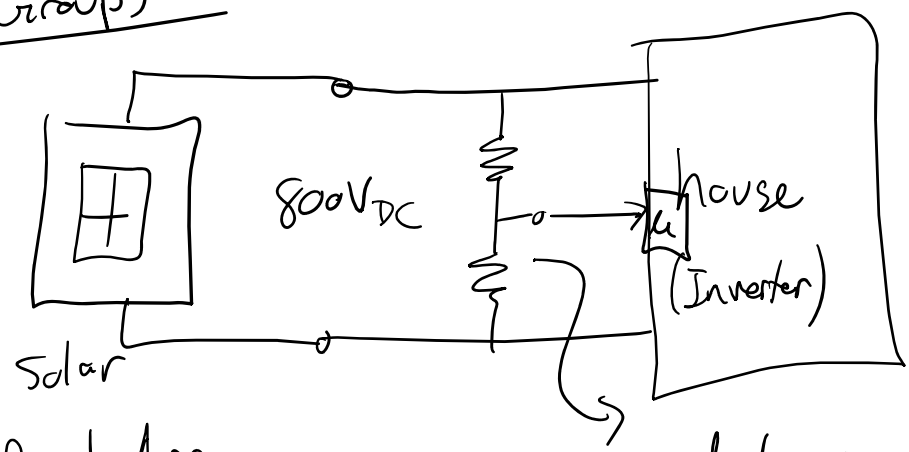
$$\boxed{R_2 = 840\Omega}$$

$$V_{out} = \left(\frac{840}{100k + 840}\right) 120V$$

$$V_{out} = 0.9996 V$$

$$\boxed{V_{out} = 1V}$$

Exp (Group)



need to measure the voltage using a microcontroller inside the house inverter.

- the analog port of the μ-controller is 3.4V

1) Design the Voltage divider that can measure the 800V_DC Solar Rail.

2) Calculate power dissipated in each resistor